

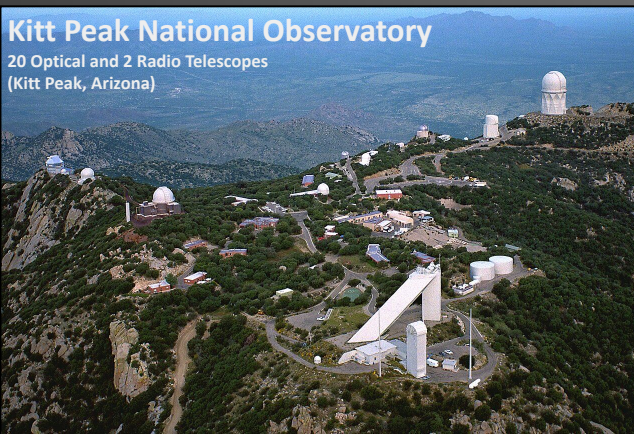
Announcements

- Next week is a day lab - we will be meeting in **LSE-S04** at **3:30 p.m.**

Intro to Telescopes

Telescopes in Arizona

- The best areas to put telescopes are those with
 - High altitudes (i.e. mountains)
 - Dry air (i.e. deserts)
 - And clear skies
- Arizona has plenty of all of those



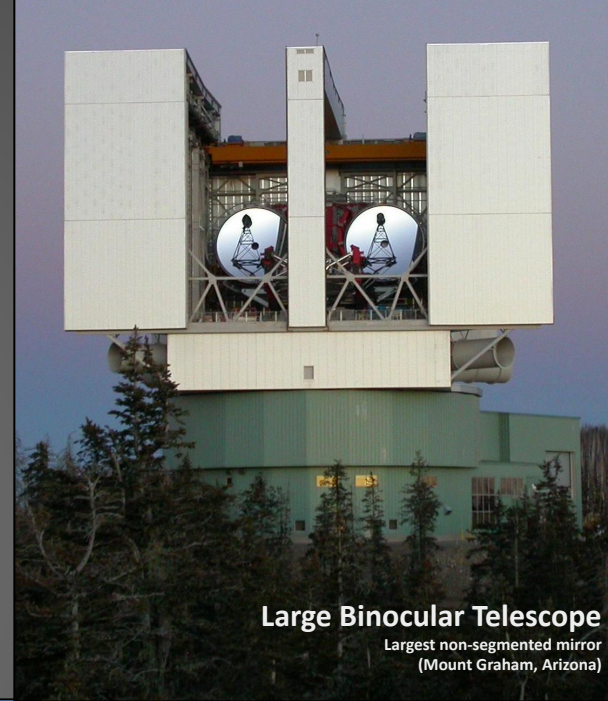
Kitt Peak National Observatory

20 Optical and 2 Radio Telescopes
(Kitt Peak, Arizona)



Lowell Observatory

Where Pluto was discovered
(Flagstaff, Arizona)



Large Binocular Telescope

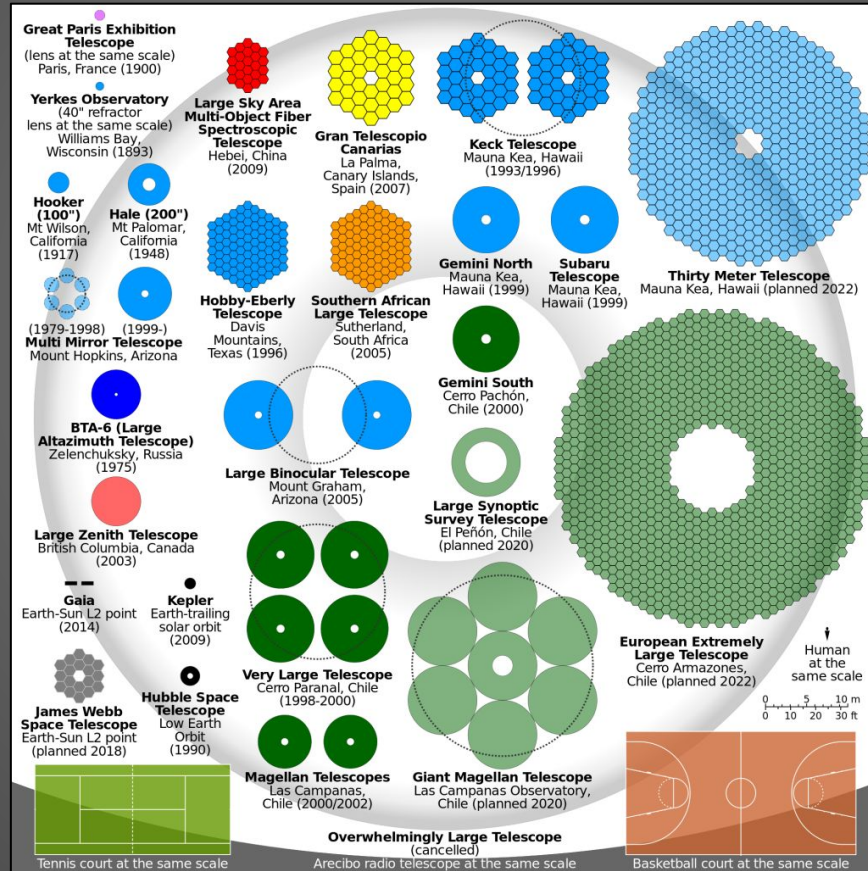
Largest non-segmented mirror
(Mount Graham, Arizona)

V.E.R.I.T.A.S Array

Very Energetic Radiation Imaging Telescope Array System
Four 12m telescopes separated by 100m
(Mount Hopkins, Arizona)

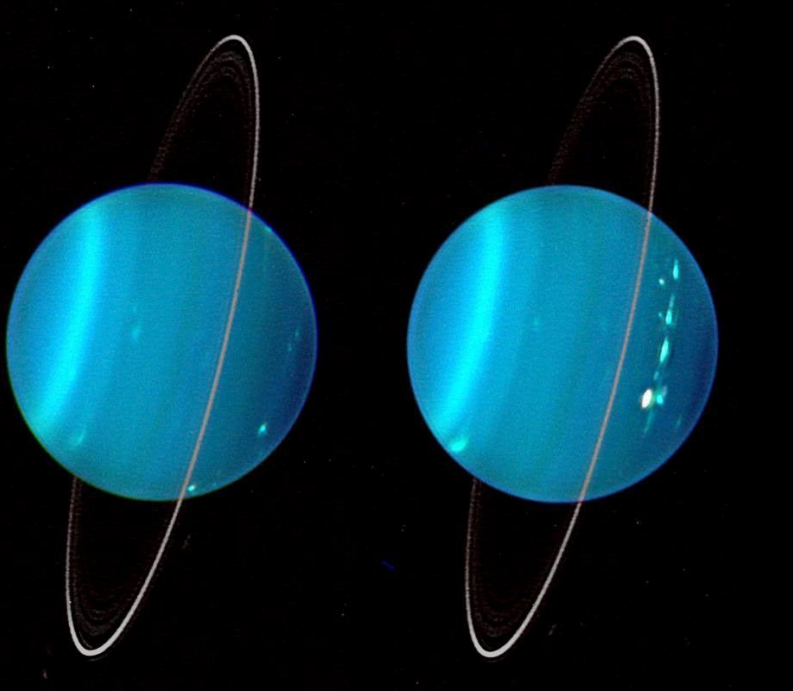


And Around the World



And Around the World

- Hawaii (Mauna Kea)
 - Keck Observatory - Two telescopes, tied for third largest mirror



And Around the World

- Spain (Canary Islands)
 - Gran Telescopio Canarias
 - Current largest single mirror in the world



M 27 - The Dumbbell Nebula
Gran Telescopio CANARIAS (GTC)



And Around the World

- Chile (Atacama Desert)
 - Vera Rubin Observatory
 - Largest Camera in the world
 - Extremely Large Telescope (ELT)
 - Under construction, will be the largest mirror when completed
 - Very Large Telescope (VLT)
 - Largest total mirror area (four mirrors combined)

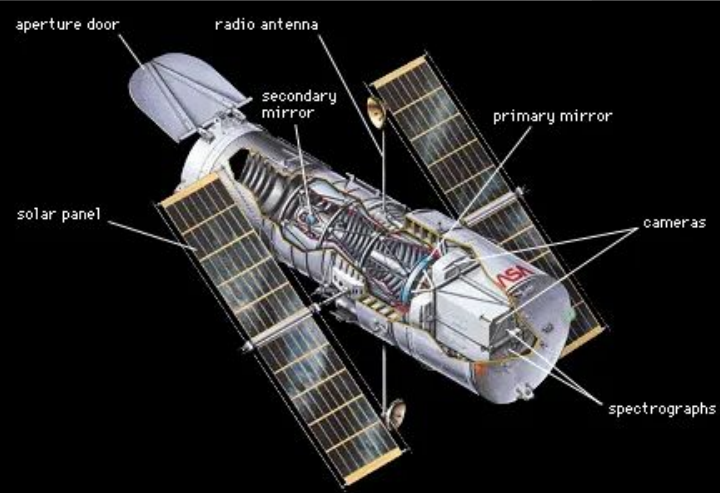


And Around the World



And Around the World

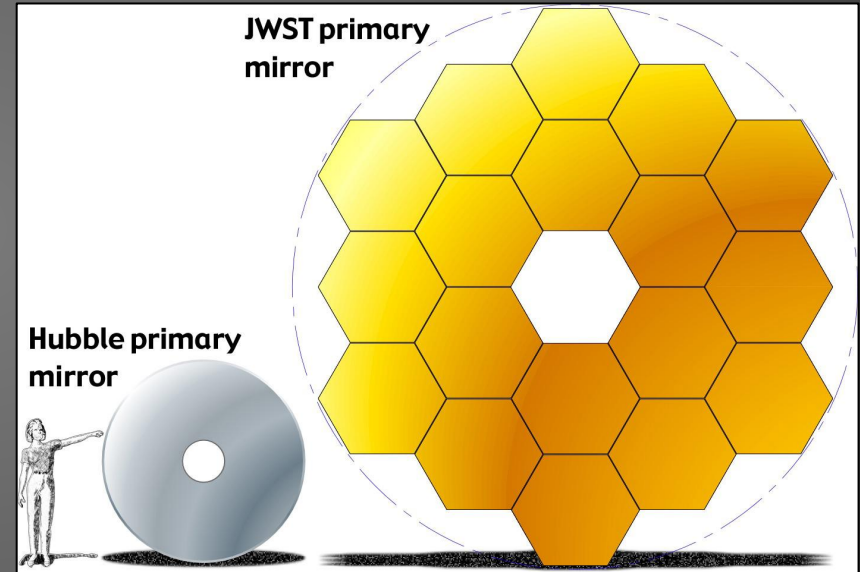
- In Space
 - Hubble (LEO)
 - James Webb (L2)
 - Nancy Grace Roman (L2)
 - Launched in August!
 - And more!



Properties of Telescopes: Light Gathering Power

- Think area
- A good analogy is a bucket collecting rainwater
 - The bigger the area, the more rain (or light) you are going to get

$$A = \pi \left(\frac{D}{2} \right)^2$$



Mirror Diameters:
Hubble: 2.4m
JWST: 6.5m

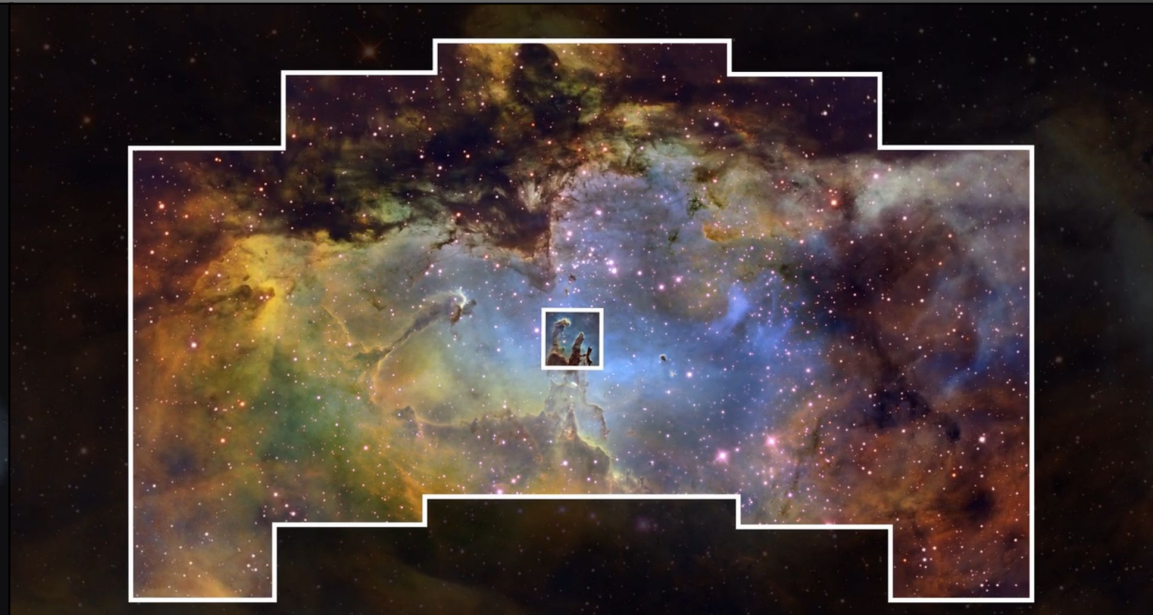
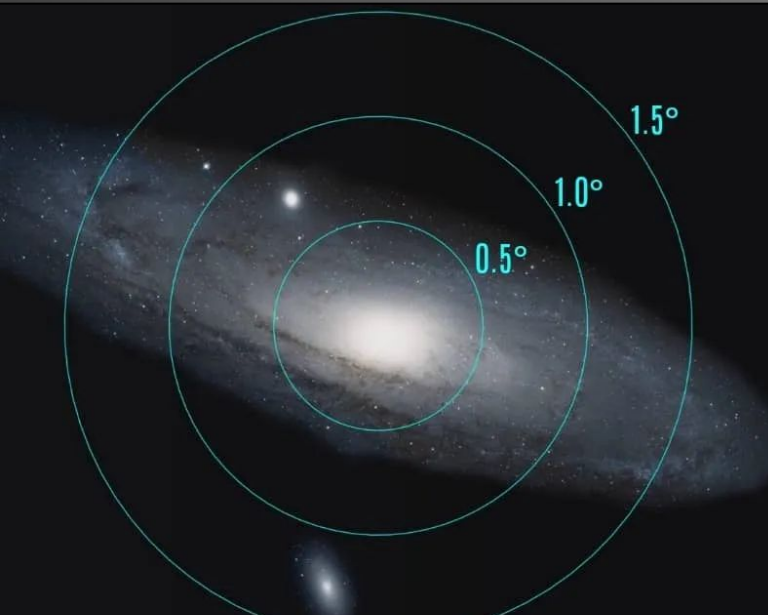
Properties of Telescopes: Focal Length and Magnification

- Focal Length is the distance between the optics, and where the image appears
- We can use the focal length to calculate the telescope's magnification
 - For ASU telescopes:
 - Telescope focal length: 2000 mm
 - Eyepiece focal length: various available, between 7 and 40 mm
 - With a 40mm eyepiece, we get a magnification of 2000/40, or 50 times magnification
 - This means that an object viewed through the telescope will appear 50 times larger than it does with the naked eye
- Note that greater magnification leads to a darker image
 - The same amount of light is getting stretched over a larger area

$$\text{MAGNIFICATION OF TELESCOPE} = \frac{\text{TELESCOPE FOCAL LENGTH}}{\text{EYEPIECE FOCAL LENGTH}}$$

Properties of Telescopes: Field of View (FoV)

- How much of the sky do you see through the telescope
 - Typically measured in angular size
- A longer focal length means a greater field of view



Properties of Telescopes: Resolving Power

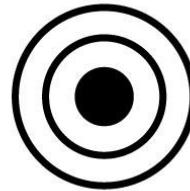
- Also referred to as “Angular Resolution”
- This is the ability to see fine detail or separate two nearby objects
- It is expressed as the minimum (angular) distance between two objects that you are still able to see them as two separate objects
- Applies to things like stars that are close together, or the smallest craters visible on the moon

$$\alpha = \frac{4.56}{D},$$

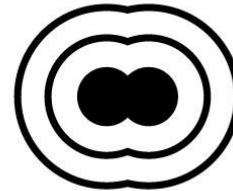
(Dawes limit for a human eye through a telescope eyepiece)

D = diameter of primary mirror in telescope in inches

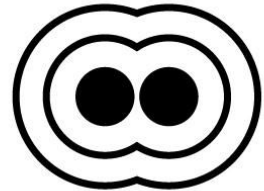
α = angular separation between objects in arcseconds



not resolved



just resolved



well resolved

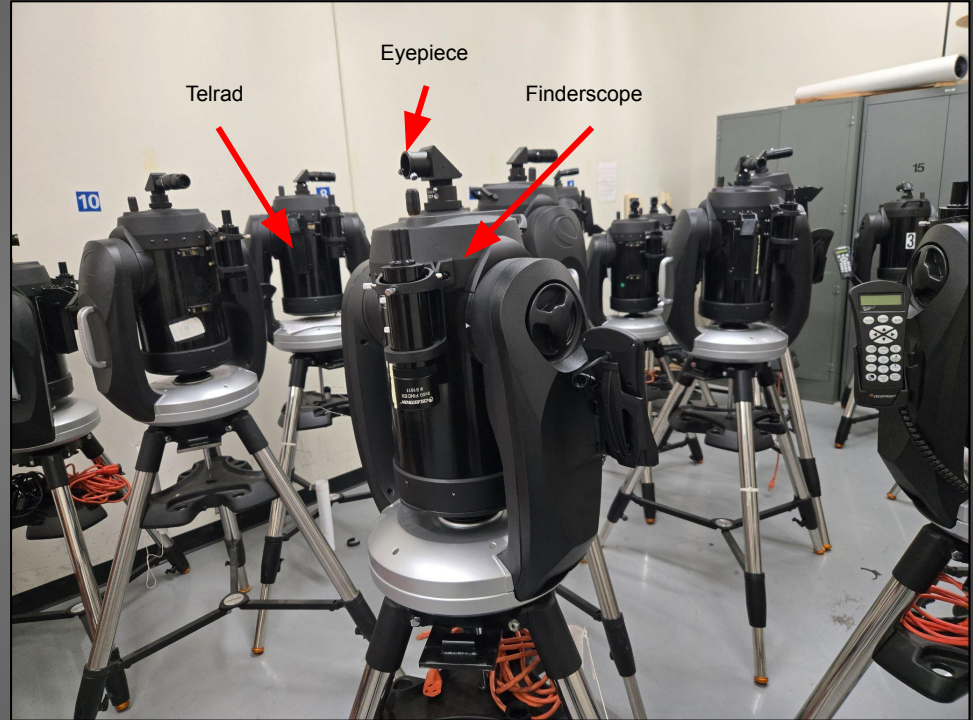
Properties of Telescopes: Scale

- How large an object appears with the telescope compared to without
- It depends on the focal length
- Typically expressed as an angle per length ($^{\circ}/\text{mm}$ or $''/\text{mm}$)
 - Sometimes, with scientific imaging from telescopes we use arcseconds per pixel ($''/\text{pixel}$)

$$\text{scale} = \frac{\text{angular size of an object on the sky}}{\text{size of the final image using the telescope}}$$

ASU's Telescopes

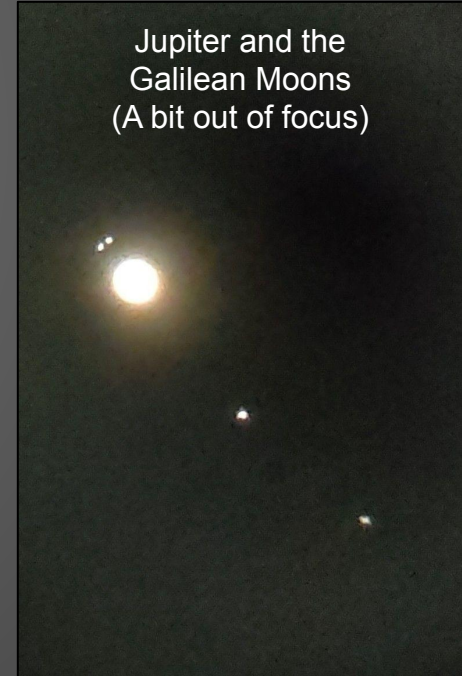
- 8 inch diameter
- 2000 mm focal length



Note: These are not in use, the orientation will be different on the roof.
Additionally, not all of the telescopes have a Telrad.

Tonight's Lab

- Skip question 1, 2, and 5 - you are not aligning the telescopes yourselves
- We will watch a couple videos to complete the table on page 7 and questions 3-4
 - Then we will go to the roof! (if weather is good)



Tonight's Lab (If weather is poor)

- We will use the Stelvision telescope simulator:
 - <https://www.stelvision.com/astro/telescope-simulator/>
- Select detailed simulation
 - input 203mm (8") for the telescope diameter and 2000mm for the focal length
 - Input eyepiece focal lengths of 40mm, 15mm and 2.5mm.
 - Choose any object(s) and sketch what you see, use the same object for all 3 sketches
- Question 12 is asking about what you think will happen, before you do it
- I will show you 3 images on the screen for you to fill in the table in after 12
 - Don't miss question 13 and the bottom of the page
- Finish the remaining questions
 - The final answer on question 14 should have units of length (i.e. inches or centimeters)
 - Question 15 is asking about the calculation in question 14, not what we are observing today
 - The "Conclusion" is a series of fill-in-the-blank questions, the options are "increase", "decrease", and "stay the same"

Roof Rules

- Bring your stuff with you
 - We will not be coming back to the classroom
- Do not enter the fenced in area
- Red flashlights only
 - No white lights except in emergencies
- Do not adjust the telescopes!
- We will be up there with one or more other classes
 - You can ask me or the other TA(s) if you have any questions
 - Take turns/work together, no need to look through the telescopes in a specific order (as long as you use the right ones for each question and/or sketch)

Tonight's Lab (On the Roof)

- Do the sketches on page 8 based on what you see in the telescopes
 - Then answer questions 6-11 based on that
- Question 12 is asking about what you think will happen, before you do it
 - Then fill in the table below it and don't miss question 13 and the bottom of the page
- Finish the remaining questions
 - The final answer on question 14 should have units of length (i.e. inches or centimeters)
 - Question 15 is asking about the calculation in question 14, not what we are observing today
 - The "Conclusion" is a series of fill-in-the-blank questions, the options are "increase", "decrease", and "stay the same"

Image Credits (In order of appearance)

- NOIRLab
 - <https://noirlab.edu/public/news/noao9809/>
- Wikipedia
 - <https://en.wikipedia.org/wiki/VERITAS>
- Wikipedia
 - <https://en.wikipedia.org/wiki/Pluto>
- Wikipedia
 - https://en.wikipedia.org/wiki/Large_Binocular_Telescope
- NASA
 - <https://asd.gsfc.nasa.gov/blueshift/index.php/2016/07/25/thirty-years-of-space-vlbi/1000/>
- NASA
 - <https://science.nasa.gov/photojournal/keck-telescope-views-of-uranus/>
- Keck Observatory
 - <https://keckobservatory.org/our-story/telescopes/>
- Astro La Palma
 - <https://lapalmastars.com/en/visit-observatory-lapalma/>
- GTC Gallery
 - <https://www.gtc.iac.es/multimedia/imageGallery.php>

Image Credits (Continued)

- National Geographic
 - <https://www.nationalgeographic.com/science/article/first-images-vera-rubin-observatory-astronomy-space>
- AURA Astronomy
 - <https://www.aura-astronomy.org/centers/nsfs-oir-lab/rubinobservatory/>
- Tenor
 - <https://tenor.com/search/tim-curry-space-gifs>
- Britannica
 - <https://www.britannica.com/topic/Hubble-Space-Telescope>
- National Geographic
 - <https://www.nationalgeographic.com/magazine/article/hubble-telescope-anniversary-10-greatest-images>
- BBC Sky at Night Magazine
 - <https://www.skyatnightmagazine.com/news/james-webb-space-telescope-pismis-24>
- Wall Street Journal
 - <https://www.wsj.com/story/everything-you-need-to-know-about-nasas-james-webb-space-telescope-b00d144e>
- Wikipedia
 - https://en.wikipedia.org/wiki/Nancy_Grace_Roman_Space_Telescope
- NASA
 - <https://science.nasa.gov/asset/webb/hubble-webb-mirror-comparison/>

Image Credits (Continued)

- The lab's own instructions
- Love the Night Sky
 - <https://lovethenightsky.com/what-is-a-field-of-view-in-astronomy-and-why-is-it-useful/>
- ESA
 - https://www.esa.int/ESA_Multimedia/Images/2023/02/Roman_s_field_of_view_compared_to_that_of_Hubble
- The lab's own instructions
- Geeks for Geeks
 - <https://www.geeksforgeeks.org/physics/resolving-power-of-microscopes-and-telescopes/>
- The lab's own instructions
- Taken by me
 - Same for the rest